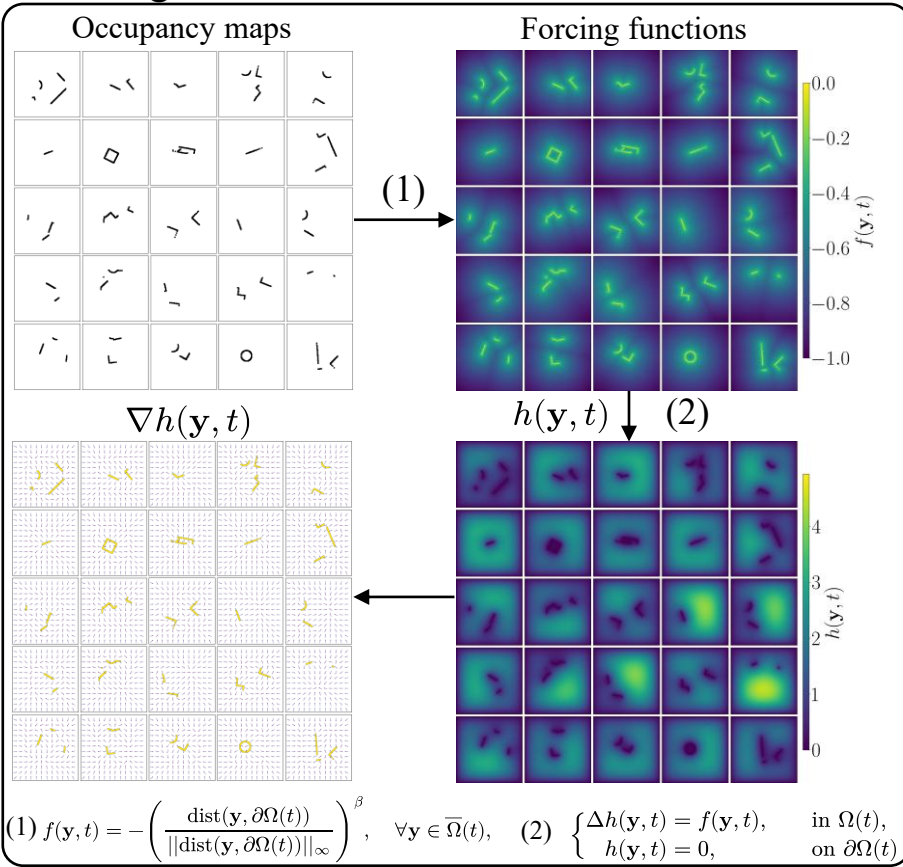


Dataset generation



Training

(< 10 mins on a single RTX 4090 GPU)

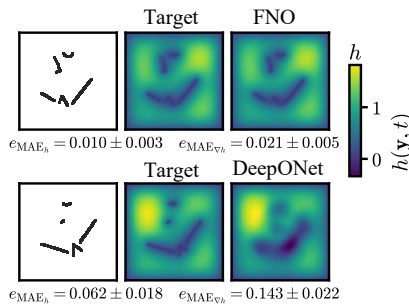
Neural Operators

- Fourier Neural Operator (FNO)
- Deep Operator Network (DeepONet)

Minimize, $\mathcal{L} = \mathcal{L}_h + \lambda_{\nabla h} \mathcal{L}_{\nabla h}$

$$\mathcal{L}_h = \frac{1}{NM} \sum_{i=1}^N \sum_{j=1}^M |\underbrace{\hat{g}_\theta(f_i(\mathbf{y}_j, t))(\mathbf{y}_j)}_{\hat{h}_i(\mathbf{y}_j, t)} - h_i(\mathbf{y}_j, t)|^2,$$

$$\mathcal{L}_{\nabla h} = \frac{1}{NM} \sum_{i=1}^N \sum_{j=1}^M \|\nabla \hat{h}_i(\mathbf{y}_j, t) - \nabla h_i(\mathbf{y}_j, t)\|^2$$



Trained model

Neural Operator

$\hat{h}(\mathbf{y}, t)$

$\nabla \hat{h}(\mathbf{y}, t)$